

SINGLE-CELL RNA-SEQ · HUMAN PBMC · 4 SAMPLES, 29,157 CELLS

Nanoplastics and the human immune response

How particle size shapes what immune cells do — 40 nm vs 200 nm carboxylated polystyrene nanoparticles, and their mixture

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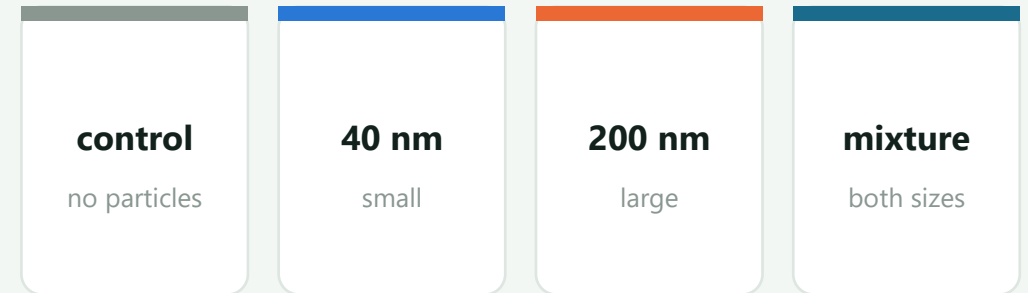
What we set out to find

Nanoplastics are small enough to enter the bloodstream, where they meet immune cells directly.

We exposed human PBMCs — peripheral blood mononuclear cells — from a single donor to carboxylated polystyrene nanoparticles.

The question:

Does particle SIZE change the immune response — and if so, what exactly changes?



Four samples · one donor · 10x Genomics single-cell RNA-seq

What we actually measured

Annotated cells per sample, after quality control:

control 5,630	40 nm 7,312
200 nm 10,793	mixture 5,204

21,862 genes measured in each. No concentration or exposure time is recorded in the metadata, so this compares sizes, not doses.

What was done

1**Quality control**

Remove dead, empty and doublet cells

2**Integration & clustering**

Remove batch effects, group similar cells

3**Cell type annotation**

Manual markers + 2 automated models

4**Composition**

How MANY of each cell type per sample

5**Differential expression**

What each cell type is DOING + pathways

6**Size-specific effects**

40 nm vs 200 nm vs the mixture

AND FOUR THINGS WE TRIED TO BREAK THE RESULT WITH

Is the response even real?

RNA leaking from broken cells lands in other droplets and imitates a response. Every pathway carries a measure of how much of it is that.

Do all the cells respond?

An average can hide two groups. We measured the inflammatory program in every individual cell instead of averaging over them.

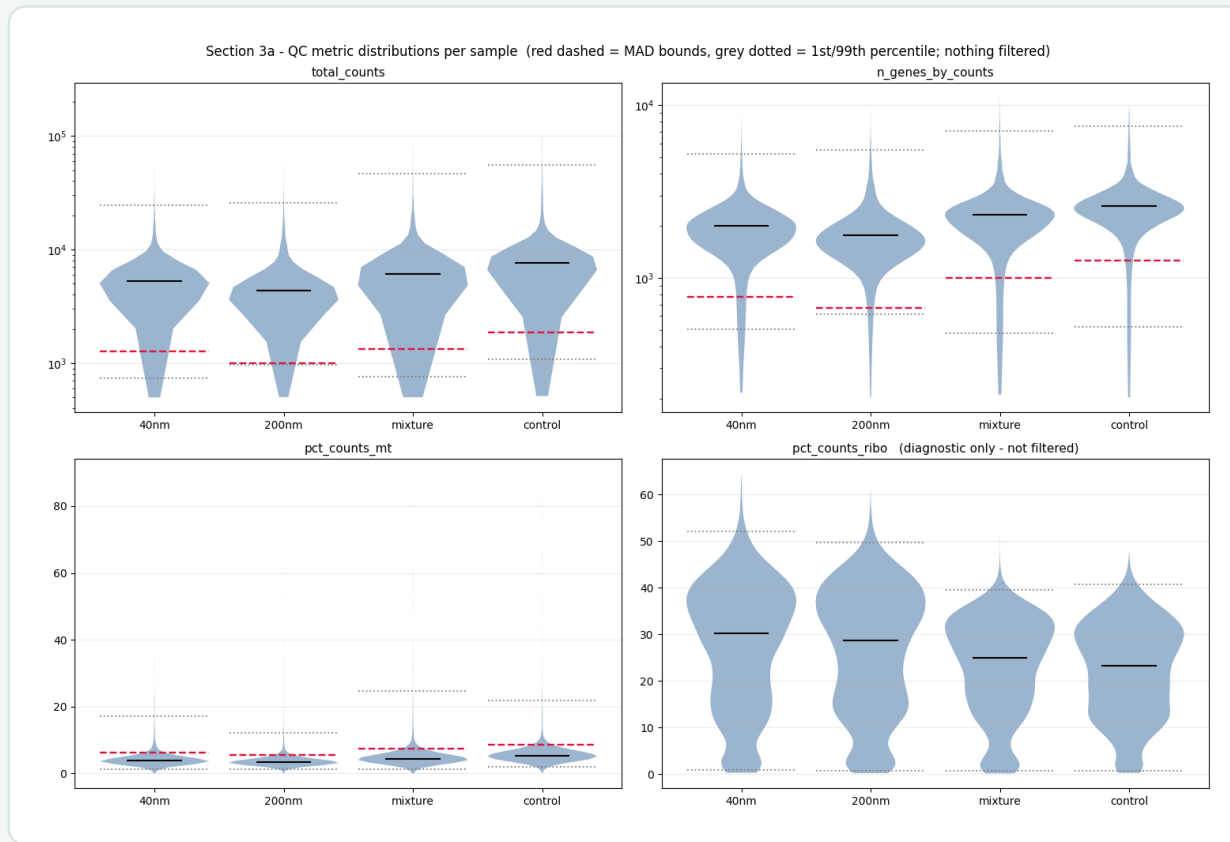
Does it survive without batch correction?

The correction step is not fully reproducible across machines, so we removed it and re-ran the whole analysis without it.

Is the mixture more than its parts?

We predicted the mixture from the two single sizes and asked what was left over that neither could account for.

Quality control — which cells are real?



34,078

loaded



31,839

doublets removed



29,157

analysed

Why the cut-offs sit where they do

The red dashed lines are the cut-offs, and they differ for every sample.

The 200 nm run was sequenced less deeply, so one fixed cut-off would have discarded far more of it — for technical reasons, not biological ones.

How they were computed

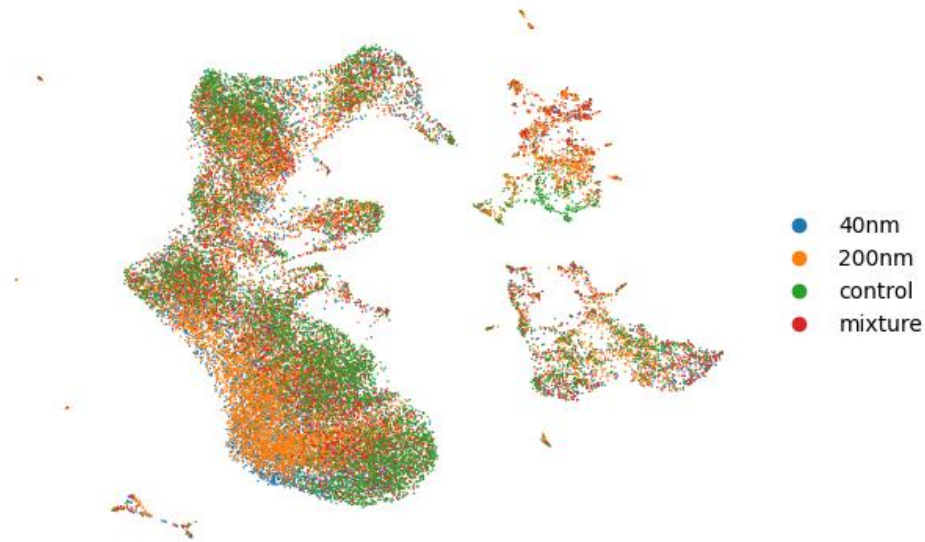
Take that sample's middle value, then how far a typical cell sits from it. A cell is cut past 5 of those distances for depth and genes, or 3 for mitochondrial content.

Every cut is one-sided, toward where cells are genuinely bad: too little RNA, or too much mitochondrial RNA.

The mitochondrial cut removes most of them.

Correcting batches, then grouping cells

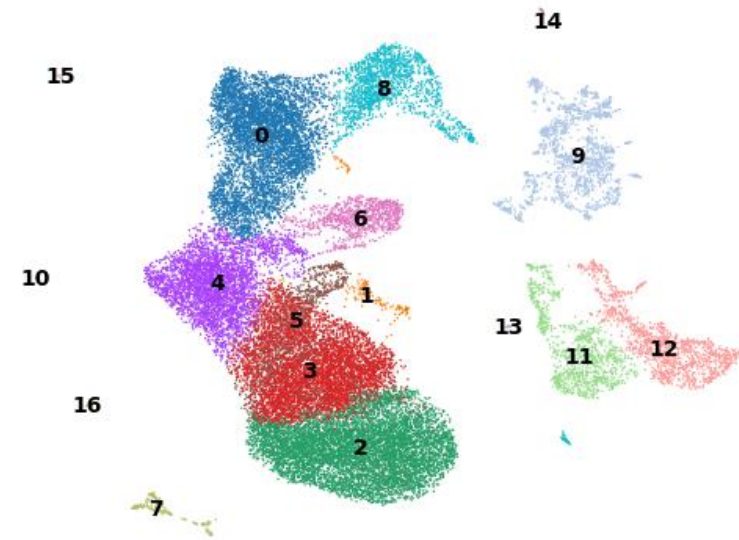
Samples after scanorama integration



Samples overlap — so batch effects are gone

Every colour is a sample. They sit on top of each other, so cells now group by BIOLOGY, not by which run they came from. Corrected with scanorama, chosen because over-correcting would erase the very size effect we are hunting.

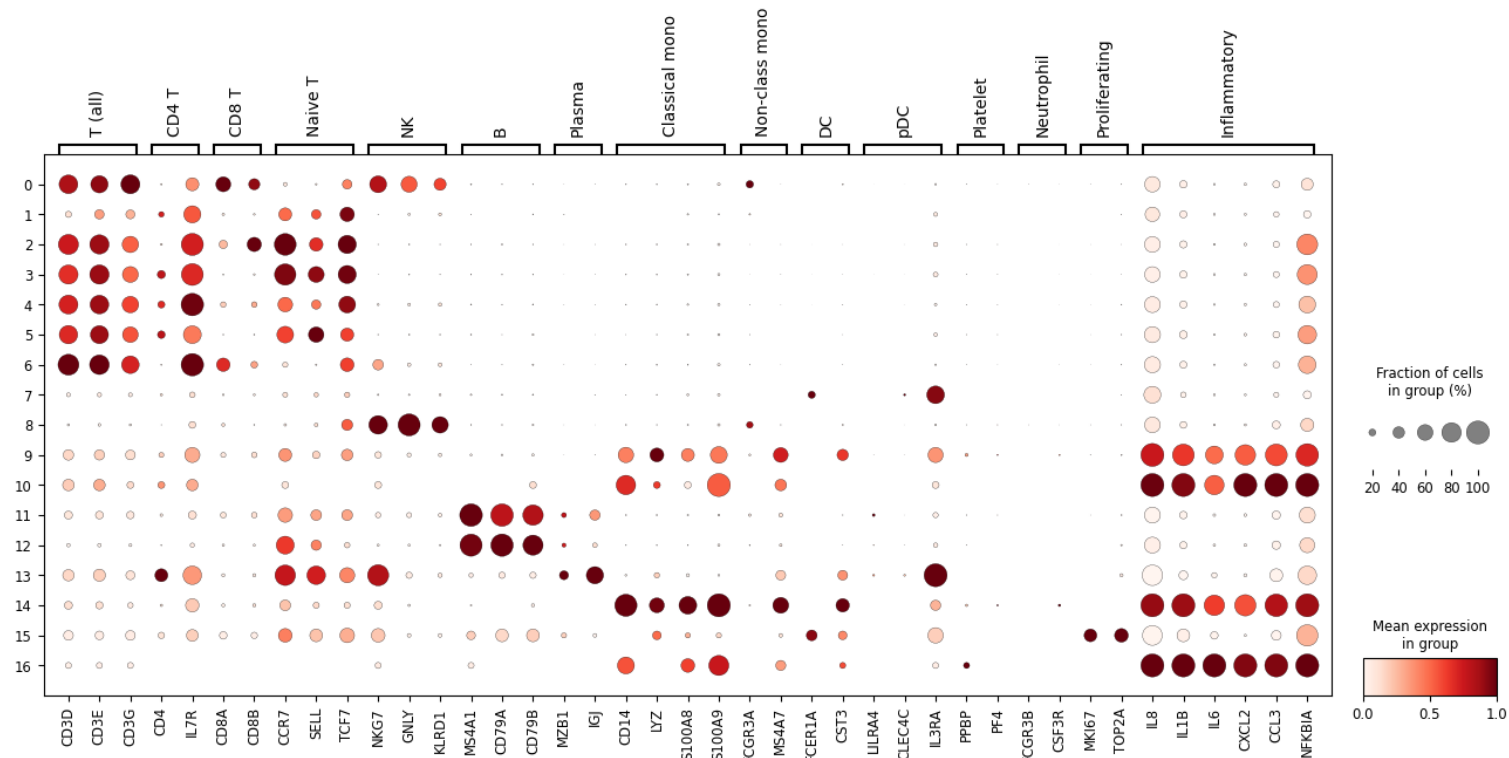
Leiden clusters (resolution 0.3)



17 clusters at resolution 0.3

Cells grouped by similarity. How finely they are split is a setting with no single correct value, so we tried ten of them and kept the one whose groups held together best when the data was repeatedly resampled.

Naming the clusters from their marker genes



The 11 cell types

CD8 T naive
7,033
B naive
1,220

CD4 T naive
5,599
B memory
1,056

CD8 T effector
5,237
Treg
911

CD4 T memory
3,579
MAIT
891

Monocyte
1,649
Basophil
156

NK
1,608

Five independent labellings, compared

Ours

manual, from
marker genes

CellTypist

logistic model,
majority voting

panhumanpy

Pan-Human Azimuth
(broad/medium/fine)

predicted. celltype

inherited with
the data

CoDi

inherited with
the data

They broadly agree

predicted.celltype agrees with our labels on 90.0% of 29,157 cells; CoDi on 89.8%. At the coarse level the picture is solid.

Compared at a shared vocabulary

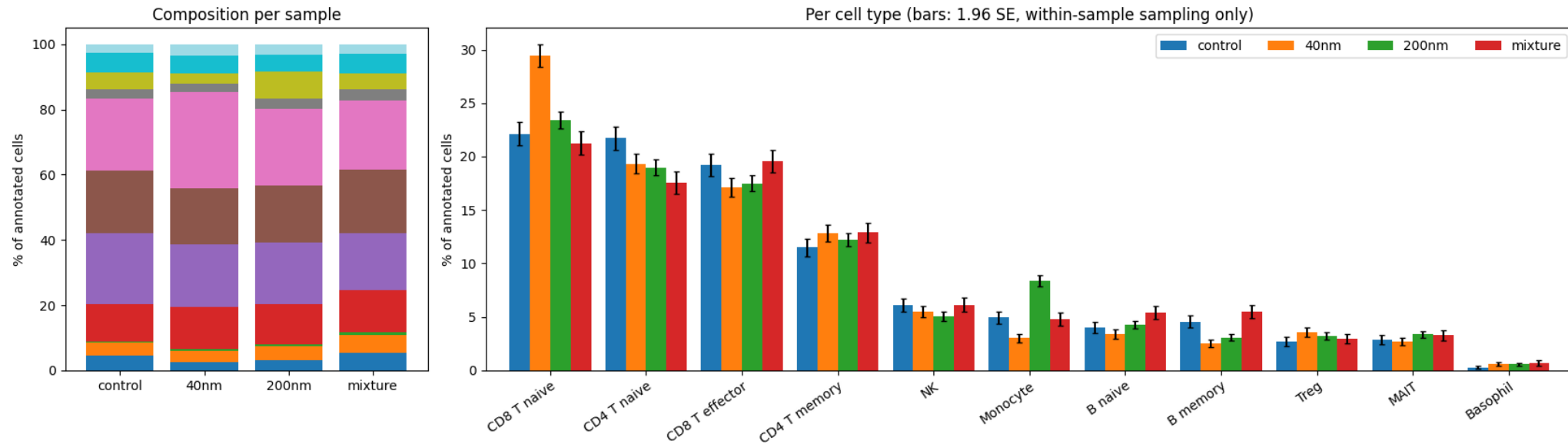
The five schemes name things differently, so each was mapped onto one common set of categories, written out explicitly rather than inferred.

And the two automated models disagree with each other — which is why they cannot be the answer.

On the 7,033-cell naive T cluster, CellTypist calls 35.7% of cells CD4 while Pan-Human Azimuth calls 87.2% — and they agree on the same individual cells only 43.9% of the time, which is below chance. Two models cannot both be right, and nothing in either output tells you which one to believe.

So the rule we adopted: automated labels are a check on our result, never the result itself. Where they conflicted with the marker genes, the genes decided, and the reasoning was written down cluster by cluster.

How many of each cell type, in each sample



Two questions, not one

How many cells of a type there are, and what those cells are doing, are separate measurements that move independently.

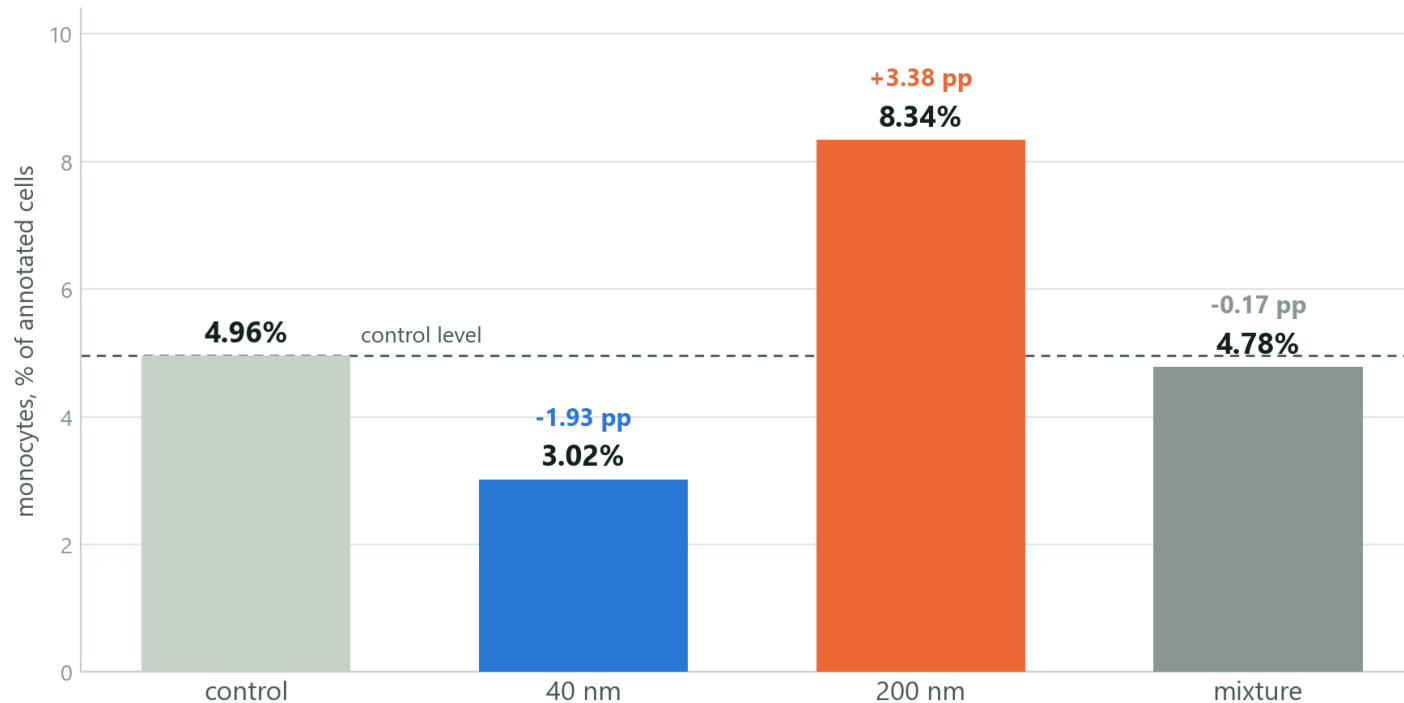
Bulk sequencing merges them: a signal can rise there simply because more of the cells carrying it are present.

What is not in these bars

218 cells in three artefact groups — doublets, and cells whose identity could not be resolved — are left out of the denominator.

So these are shares of confidently annotated cells.

Particle size changes how many monocytes there are



What this means

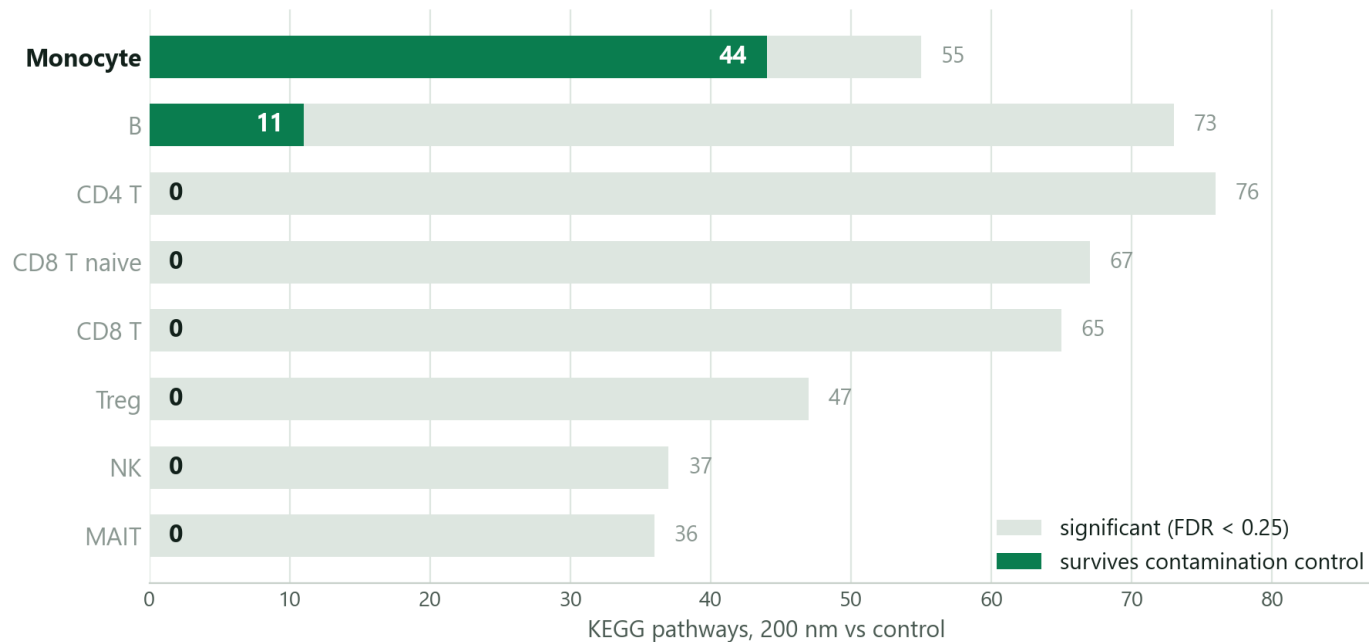
The two sizes push in **OPPOSITE** directions.

Monocytes are the phagocytes — the cells that physically engulf foreign material, and the only immune cells here that can take up a particle at all.

Fewer monocytes at 40 nm most likely means monocyte death.

The rise at 200 nm is harder to attribute: a larger share can also mean the other populations were lost.

Only one cell type actually responds



The trap we avoided

Read naively, the tables say nanoplastics activate T cells. They do not.

When a monocyte bursts, its RNA floats free and lands in every other droplet. So the same inflammatory genes appear in T cells — carried by cells holding 1–2 copies.

IL8, the worked example. In monocytes it averages 572 copies. In every other cell type it averages 1.9 to 4.4, and three-quarters of those cells hold ≤ 2 copies. That is leakage, not a response.

And it makes biological sense: only monocytes can physically engulf a plastic bead.

What the responding monocytes actually do

SWITCHED ON — an acute inflammatory program

MMP1	+6.8	breaks down tissue matrix
CSF3	+6.2	G-CSF — calls in neutrophils
CSF2	+6.2	GM-CSF — recruits myeloid cells
IL24	+6.1	cytokine
CCL20	+5.5	chemokine — recruits immune cells
PTGS2	+4.2	COX-2 — the aspirin target
IL6	+4.1	fever, acute-phase response

SWITCHED OFF — particle-catching receptors

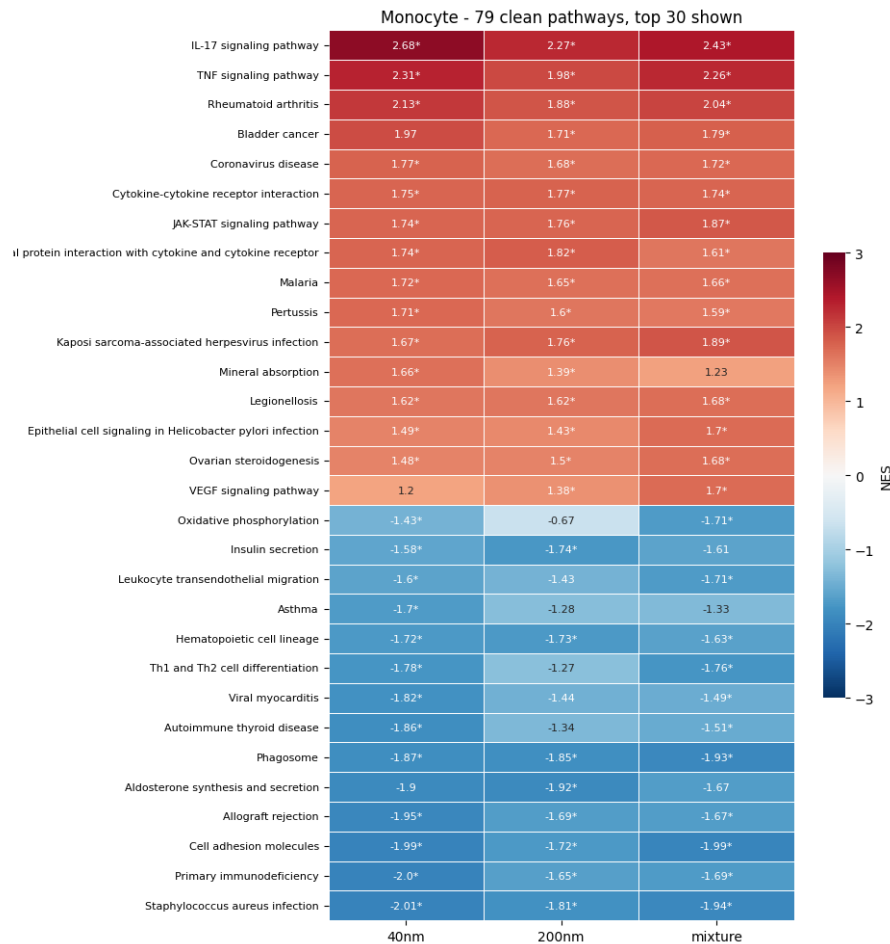
SPP1	-7.7	osteopontin
CCL1	-6.9	chemokine
GSN	-6.9	gelsolin — actin remodelling
CD36	-6.5	scavenger receptor — binds particles
GPC4	-6.2	glypican
ITGAM	-4.7	CD11b — complement receptor
CSF1	-3.3	M-CSF — differentiation signal

log2 fold change vs control at 40 nm · +6.8 means roughly 110× more of that transcript · near-identical at 200 nm

This is a textbook NF-κB acute inflammatory response.

Secrete cytokines, call in reinforcements, remodel the surrounding tissue. Note TNF itself falls while the TNF pathway rises — consistent with an immediate-early gene that has already peaked and declined, while the program it started runs on.

The same picture at pathway level



What is up

IL-17 signalling, TNF signalling, cytokine–cytokine receptor interaction, JAK-STAT — the inflammatory-signalling block, all strongly positive.

The three columns are the three exposures, and they are nearly identical. That is the point of the next slide.

The counter-intuitive one: Phagosome runs DOWN

The eating machinery scores negative in the cells that eat — consistently, in all three exposures (−1.87 / −1.85 / −1.93).

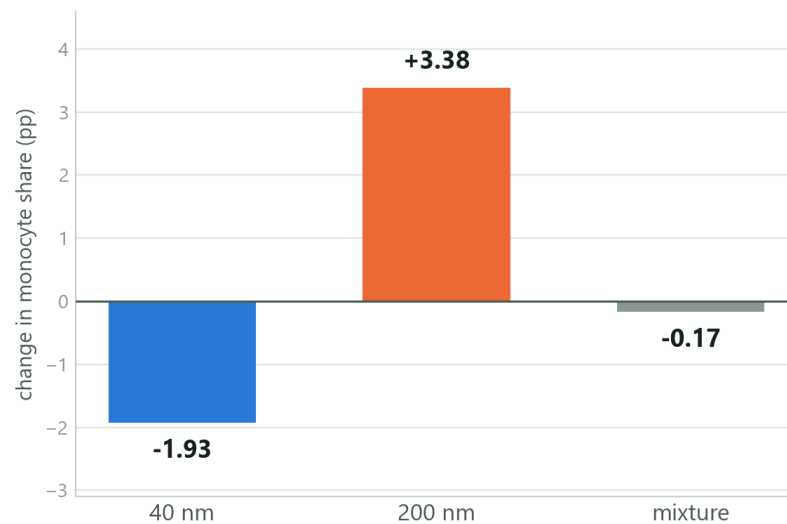
Split the gene set in two and it makes sense. The receptors that grab material from outside the cell all fall — CD36 −4.9, ITGAM −4.4, CD209 −3.8, MRC1 −2.5.

The machinery that destroys what has already been taken in rises at the same time — CD14 +3.3, THBS1 +2.6, NCF4 +1.7.

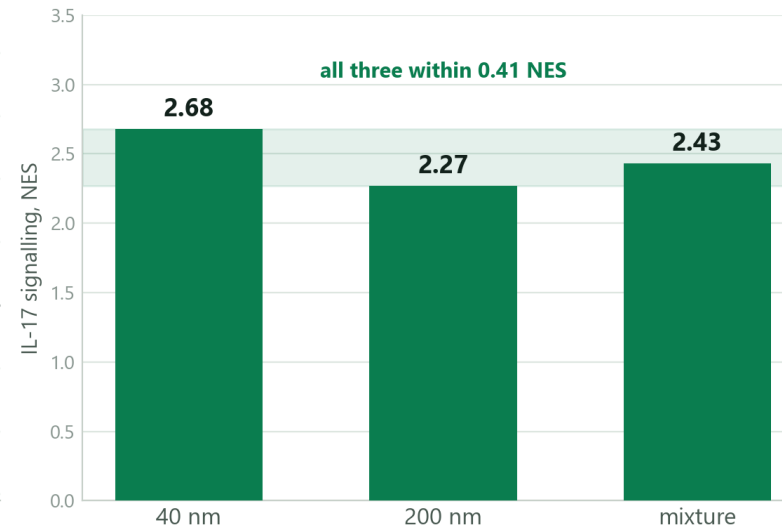
That is a cell which has already fed — not one that has stopped feeding.

The answer depends on what you measure

HOW MANY monocytes
size-specific, and directional



WHAT THEY DO
size-blind — the same program



Two correct answers

Size-specific in HOW MANY.

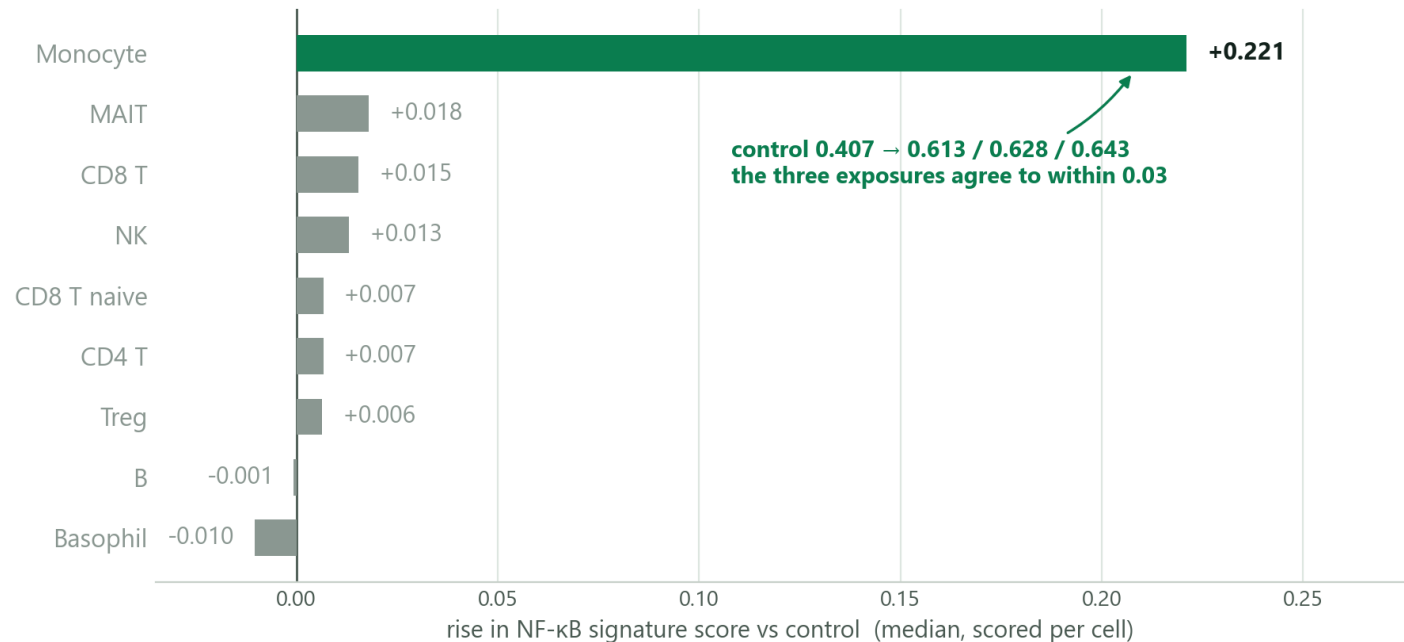
Size-blind in WHAT THEY DO.

40 nm leaves fewer monocytes and 200 nm more — but each surviving monocyte runs the same program at the same intensity.

Across 79 pathways the typical gap between exposures is a quarter of a point, on a scale running -2 to +2.7.

Not weak evidence of a difference — strong evidence of agreement.

Do all monocytes respond, or only some?



What is being measured

For each cell on its own: how strongly is it running the inflammatory program?

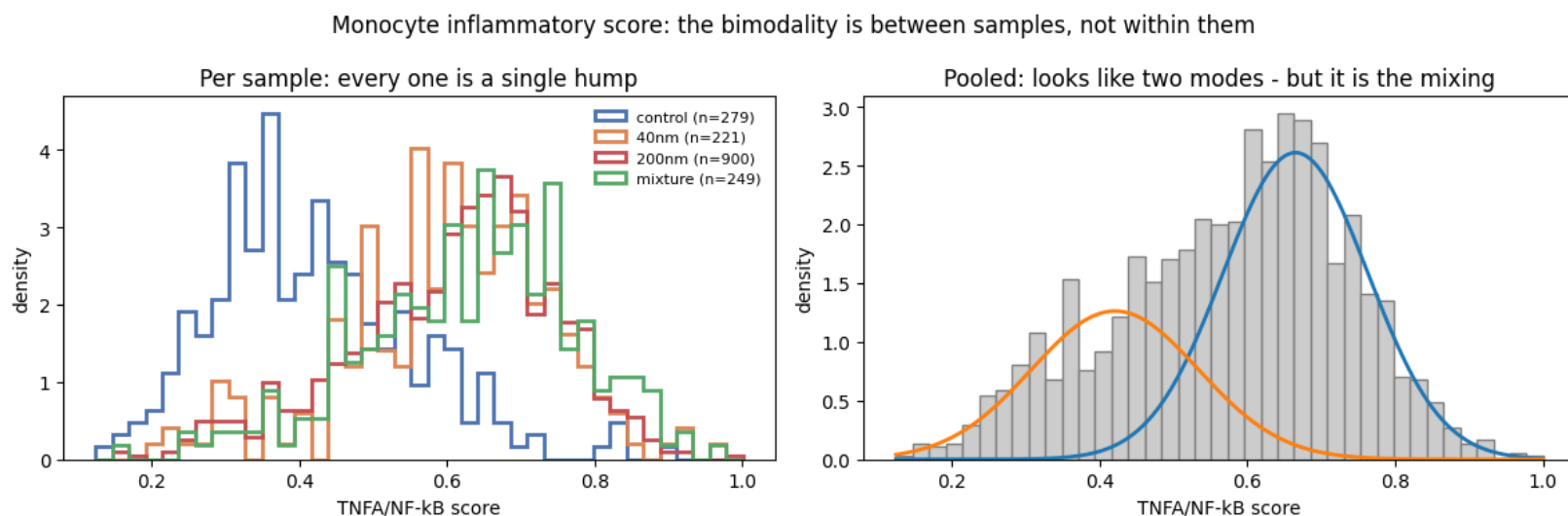
Scored against control genes matched for abundance, so a busy cell does not score high just for being busy.

Only monocytes move

Monocytes rise from 0.41 to 0.63. Every other cell type shifts by less than 0.02 — a different method reaching the same conclusion as the pathway analysis.

Six related gene sets were scored, not one — the most obvious set turns out not to contain IL8, the strongest gene in the result.

The two groups of monocytes that were not there



What you are looking at

Each curve is the spread of inflammatory scores across individual monocytes.

LEFT: one line per sample. Control sits low, the exposures sit high — and each is a single hump.

RIGHT: the same cells pooled. Now there appear to be two.

Two humps would have meant: the cells that took up a particle, and the cells that did not.

They are not. The high hump is 23% of control cells but 75–78% of exposed ones — so the two humps are control and exposed cells drawn on top of each other, which the left panel shows directly. There is no uptake-dependent subpopulation: the whole monocyte population shifts together.

Does the result need batch correction at all?

Why we had to ask

The batch-correction step is not fully reproducible. Run on a different operating system it gave slightly different output — enough to change the number of clusters from 15 to 17.

A finding that depended on it would be a finding that depended on which computer we used. So we removed it and re-ran the grouping from scratch.

The monocyte result, computed both ways

change in monocyte share vs control, percentage points

	uncorrected	corrected
40 nm	−2.01	−1.94
200 nm	+3.35	+3.34
mixture	−0.25	−0.20
the two columns never differ by more than 0.07		

The cell types come back anyway

Without correction the monocytes still form one clean group: every monocyte lands in it, and 99% of what is in it are monocytes. NK and Treg behave the same way.

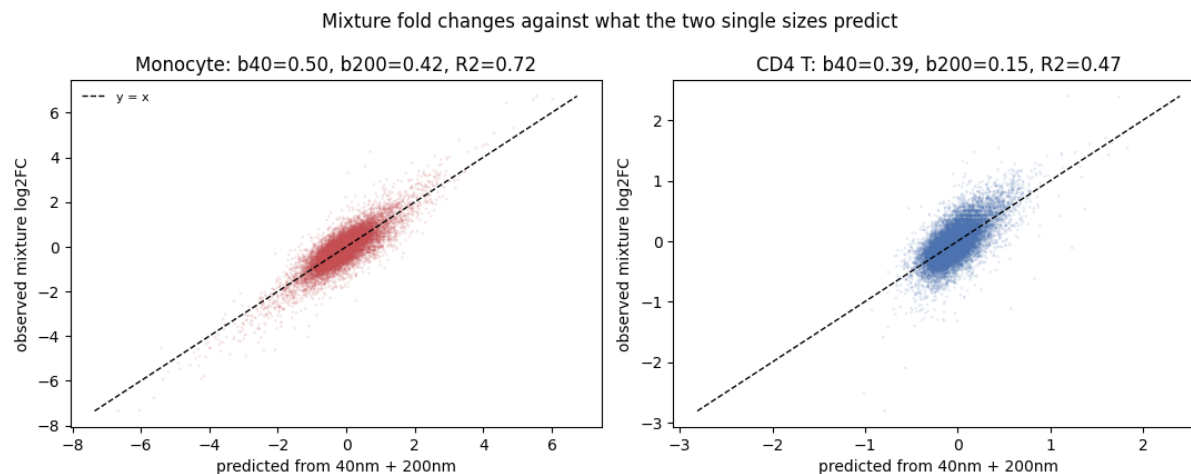
There are 11 groups instead of 17, and the two clusterings agree on 87% of the pairwise structure — so it is a coarser version of the same picture, not a different one.

What correction actually buys

One thing, and it is not the headline. Without correction the two B cell types — memory and naive — collapse into a single group.

Everything the conclusions rest on is already there before any correction is applied.

Is the mixture more than the sum of its parts?



In monocytes, the mixture is a blend

Regressing the mixture's per-gene fold changes on the two single sizes gives $b_{40} = 0.50$, $b_{200} = 0.42$ — summing to 0.93 at $R^2 0.72$, much the best fit of any cell type. Two sizes together produce no response neither produces alone.

Why the number alone is misleading

A prediction always falls short when the things doing the predicting are themselves noisy. So a score below perfect is expected even for a pure blend — there is no fixed number it should hit.

So we ran the same test two more times, each time predicting a different sample. That gives a baseline for what a perfectly ordinary sample scores here.

The result this check prevented

In lymphocytes the mixture scored low. But the other two samples score high on the same test, so that low score belongs to the mixture, not to the data.

The reason is sequencing depth: the mixture sits furthest from the others, and with no real signal, depth is most of what gets measured.

What we found

- 1 Nanoplastics trigger a real, coherent inflammatory response — in monocytes, and in no other immune cell type.**
The cell type that responds is the only one that can physically take up a particle. Everything that looked like a T cell response was ambient contamination.
- 2 Particle size controls HOW MANY monocytes there are.**
40 nm reduces them by 1.93 pp; 200 nm increases them by 3.38 pp — opposite directions from chemically identical material.
- 3 Particle size does NOT control what those monocytes do.**
The same NF- κ B program at the same intensity, at both sizes and in the mixture — confirmed independently by per-cell scoring (§16) and by mixture additivity (§18).
- 4 So “is the effect size-specific?” depends on which question you ask.**
Size-specific in composition, size-blind in expression. Reporting either alone would have been misleading — and the disagreement is itself the finding.

Governing limitation: one donor, one sample per condition. No biological replication, and sequencing depth is confounded with exposure.

**Same particles, different sizes:
the surviving cells behave the same,
but how many survive does not.**

Thank you